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In [3]: # Import libraries
import pandas as pd
import nltk
import string
from nltk.corpus import stopwords
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

# Download stopwords
nltk.download('stopwords')

# Load dataset (download from Kaggle and place in same folder)
df = pd.read_csv(r"B:\NLP Practicals 2026\IMDB Dataset.csv")

# Display first rows
print(df.head())

# ----- PREPROCESSING -----

stop_words = set(stopwords.words('english'))

def preprocess(text):
    # Lowercase
    text = text.lower()

    # Remove punctuation
    text = "".join([char for char in text if char not in string.punctuation])

    # Remove stopwords
    words = text.split()
    words = [word for word in words if word not in stop_words]

    return " ".join(words)

# Apply preprocessing
df['clean_review'] = df['review'].apply(preprocess)

# Convert labels to numeric
df['sentiment'] = df['sentiment'].map({'positive': 1, 'negative': 0})

# ----- TRAIN TEST SPLIT -----
X_train, X_test, y_train, y_test = train_test_split(
    df['clean_review'], df['sentiment'], test_size=0.2, random_state=42
)

# ----- TF-IDF -----
vectorizer = TfidfVectorizer(max_features=5000)

X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

# ----- MODEL -----
model = LogisticRegression()

model.fit(X_train_tfidf, y_train)

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# ----- PREDICTION -----
y_pred = model.predict(X_test_tfidf)

# ----- EVALUATION -----
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
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[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

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                                review sentiment
0  One of the other reviewers has mentioned that ... positive
1  A wonderful little production. <br /><br />The... positive
2  I thought this was a wonderful way to spend ti... positive
3  Basically there's a family where a little boy ... negative
4  Petter Mattei's "Love in the Time of Money" is... positive
Accuracy: 0.8891
```

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.87	0.89	4961
1	0.88	0.90	0.89	5039
accuracy			0.89	10000
macro avg	0.89	0.89	0.89	10000
weighted avg	0.89	0.89	0.89	10000

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